Abivan Vijay

Robotics, embedded hardware and web applications

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PROFILE

Designs and builds ESP32 hardware, autonomous robots and full-stack web applications, from sensors and firmware to the web interface, and takes them to national competitions, hackathons and expos.

HIGHLIGHTS

Vadiva, Velammal International School, Ponneri: winners in Robo Soccer, Robo Retrieve and Drone Race; third place in Robo Race; Innovation Contest: top 25 of 100 teams — 23–25 Oct 2026

Robotics for Good Youth Challenge (ITU): selected for the national round (Senior category), reached the semi-finals, Delhi — Sep–Oct 2026

PEC Hacks 4.0: Consolation Prize, Vault (team lead) — Aug 2026

School Shark Tank-style pitch competition: presented Volt, representing my school branch — Aug 2026

Young Environmental Scientist 2026 (IWMA with IIT Madras): participant, Sewer Gas Detector — Aug 2026

National Robotics Championship: qualified with CRASH at the SRM zonals; Delhi round upcoming — Jul 2026

School-level expo: Winner, FreshSense (team lead) — Feb 2026

RoboWunder International Competition: participant at the Regionals, FreshSense (team lead) — Jan 2026

SRM University, Ramapuram expo: participant, Smart Guard — Nov 2025

Upcoming: **National Robotics Championship 2026**, Delhi: drone soccer and innovation contest — 20–21 Nov 2026

A selection of recent highlights; more on abivan-dev.vercel.app.

PROJECTS

CRASH: Chennai Road Accident Safety Hub *Team project* — JavaScript, Python

Web app that maps road accidents across Greater Chennai, ranks junctions by a severity-weighted risk score and recommends interventions, with analytics dashboards, Monte Carlo simulation and an AI data assistant.

Outcome: [TODO: number of junctions / accident records covered] · Link: [TODO: GitHub/demo link]

Robotics for Good: Public-Health Response Robot *Team project* — LEGO SPIKE Prime

Fully autonomous robot that runs 2-minute simulated emergency-response missions: moving samples, closing a quarantine zone, delivering medical kits and colour-sorting patients.

Outcome: [TODO: mission score or tasks completed] · Link: [TODO: GitHub/demo link]

Vault: Cold-Chain Integrity Console *Team lead* — TypeScript, ESP32

Monitors a vaccine shipment's cold chain and records every event to a hash-chained ledger that can be verified afterwards.

Outcome: [TODO: e.g. number of events logged or verification time] · Link: [TODO: GitHub/demo link]

FreshSense: Food Spoilage Detection *Team lead, 3 members* — ESP32, Blynk IoT

ESP32 conveyor system that classifies food as FRESH or ROTTEN using gas sensors and a self-calibrated baseline, with remote monitoring through Blynk IoT.

Outcome: [TODO: classification accuracy or samples tested] · Link: [TODO: GitHub/demo link]

Also built: Sewer Gas Detector (ESP32 wearable for sanitation workers, ESP-NOW), Volt (peer-to-peer solar energy ledger, Open Energy Challenge 2026), Smart Guard (team lead, 3 members), WriteWise AI (Flask + Claude), Landmine Detection Shoe (prototype), RFID Door Lock.

SKILLS

Hardware and IoT: Arduino, ESP32, LEGO SPIKE Prime;
gas, climate and RFID sensors; displays, servos; Wi-Fi, ESP-NOW, Blynk IoT

Languages: JavaScript, TypeScript, Python, Arduino C++, HTML, CSS

Backends and data: Node.js, Express, FastAPI, Flask; MongoDB, Supabase; REST APIs, Claude API, Resend

Web: analytics dashboards and maps; canvas and scroll-driven animation

Techniques: sensor calibration; hash-chained ledgers; Monte Carlo simulation

Tools: Claude Code, Codex; Git, GitHub; Render, Vercel

EDUCATION

Velammal Academy, Nolambur, Chennai: Grade 9, CBSE (age 14) — 2026–27